

Direct-to-Device Non-Terrestrial Communications Ensuring Interference-Free GSO Coexistence

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Abstract—The non-geostationary orbit (NGSO) satellite communication ecosystem is rapidly expanding as new players, innovative applications and services emerge. However, coexistence between NGSO and geostationary orbit (GSO) satellite systems introduces significant challenges in managing co-channel downlink interference. In this paper, we present a novel dynamic user-to-satellite association and handover minimization that ensures interference-free coexistence with GSO systems while maintaining quality of service (QoS) for NGSO users. A significant benefit of our method is its ability to protect GSO systems from interference without requiring coordination or knowledge of their receiver locations. This is achieved by the NGSO system independently identifying and avoiding interference-prone regions through optimized user-to-satellite allocation and power control. In addition, we propose a codebook-based beam steering (CBS) method that significantly reduces the complexity and cost of NGSO user terminals. The proposed CBS approach achieves three-dimensional beam coverage by discretizing the azimuth and elevation directions while avoiding the computational burden of real-time beam steering. Simulation results demonstrate that our approach significantly reduces handover rates, maintains the required signal-to-interference-plus-noise ratio (SINR) level, and ensures robust service continuity.

Index Terms—GSO-NGSO satellite coexistence, user association, handover minimization, interference avoidance, codebook beamforming, direct-to-device communication.

I. INTRODUCTION

THE integration of ultra-dense non-geostationary orbit (NGSO) satellite constellations and terrestrial networks is envisioned as a key architecture in sixth-generation (6G) networks to enable seamless connectivity and high-speed data services [1], [2], [3]. By overcoming terrain limitations and extending coverage beyond traditional infrastructure, satellite networks play a crucial role in achieving global wireless access, ensuring reliable communication across diverse environments. While mega-constellations offer significant benefits,

such as improved broadband access and greater coverage, they also introduce new challenges [4]. With more satellites operating in shared frequency bands, the risk of interference with geostationary satellite orbit (GSO) systems is escalating. The growing overlapping footprints of NGSO and GSO systems increases the probability of co-channel downlink interference [5], threatening the operational reliability of GSO systems, which have long held priority in spectrum usage. To mitigate these interference risks, the International Telecommunication Union (ITU) has been developing regulatory provisions for NGSO systems, aimed at safeguarding the GSO networks. The outcomes of world radiocommunication conference's (WRC-23) highlight the ongoing commitment to ensuring that the expansion of NGSO constellations does not compromise the integrity and reliability of established GSO services [6]. More specifically, regions near the equator face a higher risk of interference due to frequent line-of-sight alignments between NGSO and GSO satellites. At the same time, these areas have a lower density of NGSO satellites compared to higher latitudes [7], limiting handover opportunities and making user association more challenging. As LEO satellites are only visible to ground users for a limited time, effective user-to-satellite association and timely handovers are crucial to maintaining connectivity. Unlike terrestrial networks, where base stations are stationary, the dynamic topology of satellite constellations necessitates frequent handovers as satellites move relative to the Earth. An effective handover algorithm in satellite communication must minimize unnecessary handovers while ensuring handovers are triggered promptly when needed. In particular, excessive handovers lead to increased signaling overhead and computational burden, disrupting user associations and reducing system efficiency. On the other hand, infrequent handovers can result in user disconnections or degraded service as satellites move out of range, compromising quality of service (QoS).

In addition, satellites are designed with the fundamental goal of providing global coverage, enabling connectivity to users around the world. Achieving this vision necessitates the development of cost-effective user terminal antenna receivers to ensure accessibility and scalability for widespread deployment, particularly in the emerging satellite direct-to-device (D2D) market [8]. LEO satellites traverse the sky rapidly, which requires user terminals to continuously track these moving targets to maintain a stable connection. Tracking methods typically rely on either mechanically steered parabolic antennas or complex phased array systems. Mechanically steered antennas are expensive, heavy in weight, and in size, thus limiting their application to very specific markets. On the other

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hand, phased array antennas provide more flexible beam steering capabilities. However, determining optimal beamforming coefficients often requires computationally intensive algorithms, which become increasingly complex as the number of antenna elements grows. In the context of satellite user terminals, we hypothesize that frequent beam reconfiguration should be avoided as long as the QoS remains acceptable. Only when QoS degrades should the beam steering be adjusted, either to maintain the connection with the same satellite or to switch to another satellite. This strategy is expected to simplify terminal operations and reduce system overhead.

A. Related Work

Efficient spectrum utilization and interference management are critical challenges in NGSO-GSO satellite networks. Studies have mostly focused on interference avoidance techniques to ensure coexistence between LEO satellites and GSO systems. Approaches such as arc avoidance [9], look-aside [10], power control mechanisms [11], beam hopping [12], and beamforming strategies [13] have been proposed to mitigate co-channel interference events. However, these works primarily focus on spectrum-sharing strategies within a static framework, assuming prior knowledge of GSO user terminal locations, beam patterns, or transmission parameters, and restricting protection to those specific points rather than providing a global protection mechanism independent of GSO-side information. Moreover, the probability of occurrence of in-line event, where a LEO satellite crosses the path between a GSO ground station and its intended GSO satellite, is analyzed in [14]. They further define areas for LEO satellites as forbidden zones, and argue that with beam steering the interference can be mitigated. In [15], channel allocation is analyzed within a hybrid LEO-GSO satellite system. However, the study assumes a predefined protection radius around GSO ground users, within which no LEO ground stations are placed, thereby avoiding interference by design. This does not reflect real-world scenarios, where LEO terminals may be distributed unpredictably, making interference management significantly more complex.

User-to-satellite association ensures that users connect to the most suitable satellite for reliable communication. Different user association metrics including Euclidean distance, average received power, and random selection are considered in [16]. A multi-objective user association problem, considering coverage duration, satellite load balancing, and capacity utilization for LEO satellite systems, is examined in [7]. This problem is solved sequentially over a set of discrete time segments and the satellites and users are considered static in the optimization. User association is also investigated in space-air-ground integrated networks from an energy efficiency perspective [17]. In [18] beam direction control and resource allocation for multi-beam LEO satellites is considered, where the objective is to improve users sum rate. The problem is divided into subproblems and solved with matching theory. User scheduling is also studied for a multiple LEO satellite in [19]. A heuristic user scheduling algorithm is applied to determine the links between satellites and ground users. However, existing studies do not explicitly consider the interplay between user association and NGSO-GSO interference management.

Ultra-dense constellations, combined with the high mobility of LEO satellites, significantly increase the complexity of handover management. Conventional single attribute conditional handover methods, such as RSS-based and location-based handover strategies [20], often result in excessive handovers, leading to signaling overhead and service disruptions. A multi-connectivity handover approach for LEO satellite networks is proposed in [21], enabling seamless transitions between serving and target satellites. It incorporates a location-based conditional handover that allows users to maintain connections with two satellites in the overlapping coverage areas, enhancing connection stability. The authors in [22] study a multi-criteria handover strategy for LEO satellites, considering factors such as minimum time visibility, minimum elevation angle, and user demand. Similarly, [23] proposes a handover metric that evaluates candidate satellites based on transmission link quality, satellite load balancing, and remaining visible time, selecting the satellite with the highest metric value for handover. Furthermore, mobility performance analysis of state-of-the-art handover procedures for 5G over LEO satellite networks with Earth-moving cells is considered in [24]. In [25] the handover performance in a space-ground integrated network is considered. An optimization problem to maximize the sum rate of users under satellite backhaul capacity limit is studied, and the handover between users and base stations is investigated by a matching algorithm to get the optimal association matrix. The group handover rate for an extensible multilayer LEO system with high-altitude platforms and terrestrial relays is studied in [26]. Handover procedures for the beam handover, inter-satellite handover and cross-layer handover are designed in this multilayer network architecture. In [27], the authors propose a decentralized handover decision making approach for LEO satellite networks, where each mobile terminal independently trains and uses a deep reinforcement learning model to optimize handovers. This approach shifts computational complexity to user terminals, increasing processing power and energy costs, while also requiring frequent satellite broadcasts and additional signaling. Unlike existing studies, our work prioritizes handover minimization to improve efficiency while ensuring service continuity and interference mitigation.

Implementing antenna arrays in real-world applications presents challenges. One consideration is the use of phase shifters, which play a crucial role in enabling beam steering and beamforming in analog phased array systems. In an ideal scenario, phase shifters with infinite precision would provide highly accurate control over the beam direction. However, in practical implementations, phase shifters typically have finite phase states due to hardware limitations and cost constraints [28]. By predefining a set of discrete beam directions, codebook-based methods mitigate the challenges associated with infinite phase states and hardware constraints, offering a more practical approach for beam steering in real-world deployments. Codebook analog beamforming is studied extensively in the literature for millimeter-wave frequencies, primarily focusing on hierarchical codebook designs for beamforming training. However, most existing codebook design algorithms are optimized for uniform linear arrays (ULAs) [29], [30], with limited focus on uniform planar arrays

(UPAs). In this work, we focus on UPAs and a low-resolution codebook design, reducing complexity while ensuring efficient beam coverage.

B. Contributions

As discussed in the previous section, existing studies on downlink NGSO-GSO coexistence often rely on static scenarios and predefined GSO user locations [9], [10], [11], [12], [13]. In contrast, our approach ensures interference-free operation of NGSO satellite constellation across the entire service region by dynamically identifying interference-prone areas and switching users to alternate satellites when in-line interference occurs. To address the latency and signaling costs of frequent handovers [24], [26], we explicitly incorporate handover minimization into the objective function, unlike previous studies that mainly provide heuristic evaluations of handover strategies or pursue sum rate maximization with handover controlled by predefined trigger thresholds [20], [21], [23], [24], [25], [26], [27]. Furthermore, handover to another satellite requires users to adapt their beamforming, a process that can be computationally complex and time-consuming. While existing works often assume ideal beam steering or rely on large-scale phased arrays, we propose a codebook-based beam steering (CBS) method that simplifies beam adaptation while ensuring efficient three-dimensional coverage. This distinction makes our approach particularly suitable for satellite user terminals, where simplicity and low computational overhead are critical. To the best of our knowledge, this is the first study to integrate interference-aware user allocation with handover optimization, providing a comprehensive solution for NGSO-GSO satellite networks coexistence. Moreover, we introduce an efficient user terminal beam steering design that reduces complexity and cost, further enhancing the practicality of our approach. The main contributions of this paper are summarized as follows:

- We propose a dynamic user-to-satellite association method for NGSO satellite systems that ensures no downlink interference with coexisting GSO systems. This is achieved by identifying in-line areas between the GSO arc and LEO satellites on Earth, allowing us to determine and avoid forbidden zones with potential co-channel interference. A key advantage of our approach is that it enables the NGSO constellation to coexist interference-free with GSO systems without making any assumptions or requiring any information about GSO terminals or their locations. Instead, it dynamically identifies and protects interference-prone regions with the movement of NGSO satellites.
- We propose a novel formulation of the user-to-satellite association problem as a handover minimization (HOMN) framework that optimizes the satellites beam power and user association. Unlike existing studies, the formulation integrates forbidden zone constraints to ensure no co-channel interference to the existing GSO systems while guarantying QoS for NGSO satellite users.
- We compare our HOMN approach with established handover strategies, including minimum elevation angle (Evmn), link quality maximization (Lnm), and sum rate

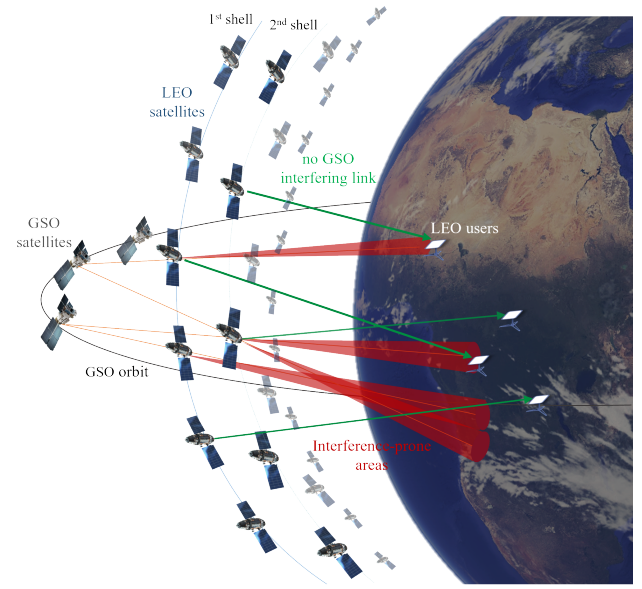


Fig. 1. Illustration of interference-prone areas and resulting user-to-satellite associations in an NGSO-GSO coexistence scenario (not to scale).

maximization (Rmx) to evaluate trade-offs in terms of handover rate, spectral efficiency, and interference to GSO systems.

- To simplify beam steering for NGSO user terminal deployments and achieve cost-effective solutions, we also propose a low-complexity, wide-beamwidth codebook-based beam steering method that efficiently covers the elevation and azimuth angle domains. A performance analysis is conducted, comparing the proposed CBS method with both a fully adaptive beam steering (ABS) model and a discrete Fourier transform (DFT)-based beam steering (DBS) scheme, thereby highlighting the trade-offs in handover frequency and received spectral efficiency.
- This study integrates user-to-satellite association, handover minimization, user terminal simplicity and GSO interference avoidance into a unified framework for NGSO satellite networks.

The rest of the paper is structured as follows: Section II introduces the system model and defines interference-prone zones and handover. In Section III, we present the user terminal codebook design and formulate the problem. Section IV details our proposed solution, followed by simulation results in Section V. Finally, we conclude the paper in Section VI.

II. SYSTEM MODEL

In this section, we define the system scenario and develop a mathematical model to characterize the interference-prone zones. We then introduce the concept of handover and discuss the downlink received signal. Table I summarizes the key notations used in this work.

A. Scenario

As shown in Fig. 1, we consider a downlink communication scenario where LEO and GSO satellites coexist and have co-channel frequency usage. We aim to prevent interference from

TABLE I
SUMMARY OF KEY NOTATIONS

Notation	Description
$\mathcal{M}$	Set of LEO satellites
$\mathcal{K}$	Set of user terminals
$\mathcal{T}$	Set of time slots
$\mathcal{G}$	Set of GSO satellites
$FZ_{k,i}^t$	Forbidden zone of satellite i at user k and time t
α	Half-angle of the forbidden zone cone
$s_{k,i}^t$	Link between user k and satellite i at time t
$h_{k,i}^t$	Channel coefficient of user k served by satellite i
$\eta_{i,k}^t$	Off-boresight transmit angle of satellite i to user k at time t
ω	User terminal receiving zenith angle
ψ	User terminal receiving azimuth angle
$G_t(\eta)$	Satellite antenna transmit gain
$G_r(\omega, \psi)$	User receiver antenna gain
$y_{k,i}^t$	Received signal at user k served by satellite i at time t
$\mathcal{HO}$	Total number of handovers
$p_{k,i}^t$	Transmit power from satellite i to user k at time t
$\gamma_{k,i}^t$	SINR of k -th user from i -th satellite
$r_{k,i}^t$	Spectral efficiency of user k in time t from satellite i
$A_e(\omega, \psi)$	Antenna element radiation pattern
$\mathbf{W}$	Set of codewords
$A(\mathbf{w}_l, \omega, \psi)$	User terminal antenna radiation pattern for codeword l
$C(\mathbf{w}_l)$	Coverage angle region for codeword l
$R_{\min}^k$	Minimum required spectral efficiency of user k
$\beta_{k,i}$	Receiving elevation angle of user k from satellite i
$\beta_{\min}$	Minimum user elevation angle
$b_{\max}$	Satellite maximum number of beams
$P_{\max}$	Satellite maximum beam power

TABLE II
LEO CONSTELLATION PARAMETERS

Parameter	Shell 1	Shell 2
Altitude	550 km	540 km
Satellites per orbit	72	72
Number of orbits	22	22
Inclination angle	53.0°	53.2°

LEO satellite beams to all potential GSO terminals, irrespective of their location or configuration. Building on the work in [14], we evaluate the coverage area of LEO satellites in regions prone to interference, which are referred to as forbidden zones, as shown in Fig. 1. Nonetheless, the system under the proposed method can operate without requiring prior knowledge of GSO user locations or the GSO satellites they are connected to. The user-to-satellite association and handover problem is addressed over a specified time frame of T time slots, each with a duration of Δt . Handover decisions are made by a ground-based network management center (NMC) that manages a subset of LEO satellites through its associated gateways. We assume that the NMC is colocated with the gateway, and that the managed subset consists of satellites visible to that gateway within the time frame T . Using known satellite ephemeris and user state information within its coverage region, each NMC can make scalable and timely handover decisions. In this study, we model a Walker-delta constellation consisting of two LEO orbital shells. Each shell consists of 1,440 satellites, and the detailed parameters of the shells, including the orbital altitude, inclination angle, and orbital arrangement, are summarized in Table II. Let M and K denote the total number of LEO satellites and user terminals, respectively. We define the set of LEO satellites as $\mathcal{M} = \{1, \dots, i, \dots, M\}$, the set of satellite user terminals as $\mathcal{K} = \{1, \dots, k, \dots, K\}$, and the set of time

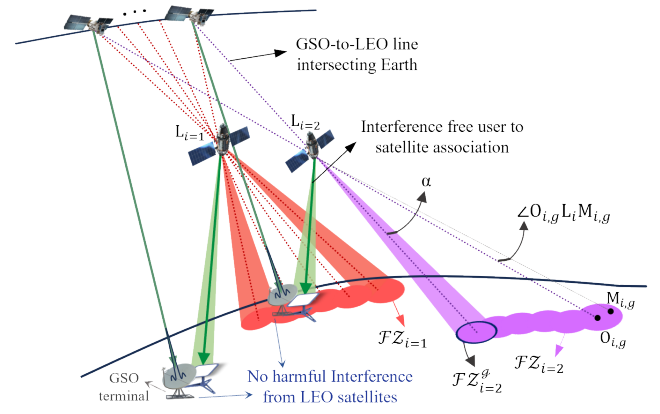


Fig. 2. Example of forbidden zones of LEO satellites determined from virtual GSO satellite positions. The illustrated LEO user lies within $FZ_{i=1}$ and therefore cannot connect to satellite $i = 1$, but is served by the second satellite. The NGSO satellites are prevented from transmitting within the high gain main-beam directions of GSO terminals, ensuring that any residual interference is received through the lower gain sidelobes of GSO antennas.

slots as $\mathcal{T} = \{1, \dots, t, \dots, T\}$. In our system model, LEO satellites support a maximum number of beams, each capable of dynamic steering. During the assigned service interval, these beams continuously track their intended users as the satellite moves [5]. The user terminals also perform beam steering to the serving satellite through two different schemes, as detailed in Section III-A.

B. Forbidden Zones

In this section, we define the forbidden zone as regions where LEO satellite transmissions must be restricted to prevent harmful interference with GSO systems. These zones correspond to in-line and near in-line geometries. To capture these conditions, for each LEO–GSO satellite pair, a three-dimensional conical exclusion region is constructed with its apex at the LEO satellite and its axis aligned with the line connecting the LEO satellite to the GSO satellite orbital position. The cone angle is determined by the LEO satellite antenna beamwidth and can be conservatively expanded to include a protection margin. The intersection of this cone with the Earth's surface defines the forbidden zone associated with that LEO satellite and GSO orbital slot. They represent geographic areas where the LEO satellite beam direction approaches the GSO receiver boresight, creating a high risk of harmful interference to potential GSO terminals. As illustrated in Fig. 2, when a LEO user falls within the forbidden zone of a particular LEO satellite, it is assigned to an alternative satellite to ensure harmful interference-free operation. As demonstrated in [14], harmful NGSO-to-GSO interference is primarily driven by in-line geometries, and a LEO satellite can avoid downlink interference to GSO earth stations by not pointing its beam toward regions where an in-line event may occur. By suppressing these dominant contributors, the aggregate interference from the remaining visible NGSO satellites remains within regulatory limits.

In the following, We use an Earth-centered, Earth-fixed (ECEF) coordinate system, with positions expressed as (x, y, z) . To determine the forbidden zones, we define the set of

GSO satellites along the orbital arc as $\mathcal{G} = \{1, \dots, g, \dots, G\}$. The g -th GSO satellite is located at coordinates (x_g^G, y_g^G, z_g^G) . Similarly, the i -th LEO satellite has coordinates (x_i^L, y_i^L, z_i^L) . The forbidden zone of a LEO satellite is then obtained by computing the intersection between its beam cone and the Earth's surface. We can consider the earth sphere, denoted by S_{Earth} with,

$$S_{Earth} = \left\{ (x, y, z) \in \mathbb{R}^3 \mid \|[x, y, z]\| = r_E \right\}, \quad (1)$$

where $\|\cdot\|$, $\mathbb{R}^3$, and r_E denote the Euclidean norm, three-dimensional real space and the Earth radius, respectively. We define the connecting line between the GSO and LEO satellite as

$$\frac{x - x_g^G}{x_g^G - x_i^L} = \frac{y - y_g^G}{y_g^G - y_i^L} = \frac{z - z_g^G}{z_g^G - z_i^L}. \quad (2)$$

If we extend the line from LEO satellite side it will hit the earth at a point we call $O_{i,g}$. This point can be calculated by the earth surface and the line equations (1) and (2). The line connecting the LEO satellite to point $O_{i,g}$ is the cones axis and the vertex is the LEO satellite. To define the forbidden zone, we consider an arbitrary point $M_{i,g}$ located inside the cone and on the Earth's surface. The cone half-angle, denoted by α , corresponds to half of the satellite's half-power beamwidth (HPBW). As a precautionary measure, a larger cone angle with additional safety margin can also be considered. The angle $\angle O_{i,g} L_i M_{i,g}$ determines whether a point lies inside the forbidden zone, and is given by

$$\angle O_{i,g} L_i M_{i,g} = \cos^{-1} \left(\frac{L_i O_{i,g} \cdot L_i M_{i,g}}{\|L_i O_{i,g}\| \|L_i M_{i,g}\|} \right). \quad (3)$$

We define the forbidden zone, $\mathcal{FZ}_i^g$, as the subset of points on the Earth's surface that fall within the cone formed by LEO satellite i and the g -th GSO satellite as

$$\mathcal{FZ}_i^g = \left\{ M_{i,g} = (x, y, z) \in \mathbb{R}^3 \mid \|[x, y, z]\| = r_E, \right. \\ \left. \angle O_{i,g} L_i M_{i,g} \leq \alpha \right\}. \quad (4)$$

Now the total forbidden zone of the i -th LEO satellite is obtained by the union of forbidden zones for each corresponding GSO location, denoted by $\mathcal{FZ}_i$ and presented as

$$\mathcal{FZ}_i = \bigcup_{g=1}^G \mathcal{FZ}_i^g. \quad (5)$$

It should be noted that the restricted interference area associated with the i -th satellite is defined at a specific point in time, as the satellite's position and the corresponding forbidden cone vary dynamically over time. In subsequent sections of this paper, the temporal index is incorporated into the equations to account for the time-dependent nature of the forbidden zones.

C. Handover and Received Signal Model

As discussed, handover and switching satellites causes delays and signaling costs, which our framework aims to

minimize. We define the binary variable $s_{k,i}^t$ denoting the link between user k and satellite i at time step t as follows,

$$s_{k,i}^t = \begin{cases} 1, & \text{if user } k \text{ is assigned to satellite } i, \\ 0, & \text{otherwise.} \end{cases} \quad (6)$$

We define another binary variable representing whether beam of satellite i at time t at user k location, falls into the satellite's forbidden zone or not. Let $FZ_{k,i}^t$ denote this variable which is defined as

$$FZ_{k,i}^t = \begin{cases} 1, & \text{if user } k \text{ is inside } \mathcal{FZ}_i, \\ 0, & \text{otherwise.} \end{cases} \quad (7)$$

We next define the total number of handovers, denoted by $\mathcal{HO}$, as

$$\mathcal{HO} = \sum_k \sum_i \sum_t |s_{k,i}^t - s_{k,i}^{t-1}| / 2, \quad (8)$$

where the division by 2 is added since at the summation over satellites, the handover to other satellite is counted twice. At the considered frequencies and elevation angles, satellite links are predominantly line-of-sight (LoS), and free-space propagation dominates the link budget, consistent with deployments where user terminals are installed to ensure obstruction-free sky visibility. Following [31], the channel coefficient, $h_{k,i \rightarrow k}^t$, that is between satellite i and user k , when i is serving user k ($i \rightarrow k$), is given by

$$h_{k,i \rightarrow k}^t = \frac{10^{(\text{Gt}(\eta_{k,i \rightarrow k}^t) \text{Gr}(\omega_{i,k}^t, \psi_{i,k}^t)) / 20}}{4\pi d_{i,k}^t / \lambda}, \quad (9)$$

where Gt indicate the satellite antenna transmit gain and $\eta_{i,k}^t$ indicates the beam off-boresight transmit angle to the user. The wavelength and the distance from user to satellite are presented by λ and $d_{i,k}^t$, respectively. The receiver antenna gain is indicated with Gr , and $\omega_{i,k}^t$ and $\psi_{i,k}^t$ present the receiving angles as shown in Fig. 3 (a). Accordingly, we define the channel coefficient between user k and satellite i which is serving another user j , by $h_{k,i \rightarrow j}^t = \frac{10^{(\text{Gt}(\eta_{k,i \rightarrow j}^t) \text{Gr}(\omega_{i,k}^t, \psi_{i,k}^t)) / 20}}{4\pi d_{i,k}^t / \lambda}$. This channel coefficient will be used to model the LEO intra-system interference.

The radiation pattern of the LEO satellite antenna is modeled according to the standard [32] as

$$\text{Gt}(\eta) = \begin{cases} \text{Gt}^{\max} - 3(\eta/\eta_b)^{\alpha_s}, & 0 \leq \eta \leq a_s \eta_b, \\ \text{Gt}^{\max} + L_N + 20 \log_{10} z_s, & a_s \eta_b < \eta \leq 0.5 b_s \eta_b, \\ \text{Gt}^{\max} + L_N, & 0.5 b_s \eta_b < \eta \leq b_s \eta_b, \\ X_s - 25 \log_{10} \eta, & b_s \eta_b < \eta \leq Y_s, \\ L_F, & Y_s < \eta \leq 90^\circ, \\ L_B, & 90^\circ < \eta \leq 180^\circ, \end{cases} \quad (10)$$

where,

$$\begin{aligned} \text{Gt}^{\max} &= 20 \log_{10}(\pi D_s / \lambda), \\ X_s &= \text{Gt}^{\max} + L_N + 25 \log(b_s \eta_b), \\ Y_s &= b_s \eta_b 10^{0.04(\text{Gt}^{\max} + L_N - L_F)}, \end{aligned}$$

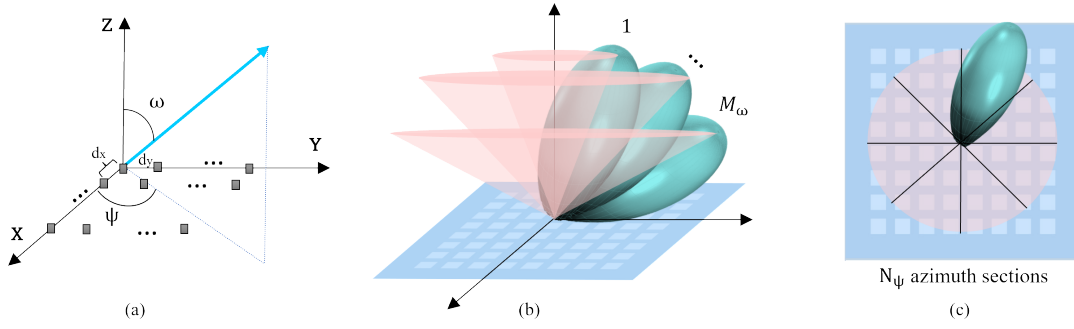


Fig. 3. Antenna coordinate and angular scan for the proposed CBS, (a) Coordinate system and angles definition. (b) Illustration of M_ω elevation layers. (c) Illustration of N_ψ azimuth subsections.

$$L_B = \max(0, 15 + L_N + 0.25G_t^{\max} + 5 \log z_s), \quad (11)$$

and η represents the off-boresight angle in degrees, η_b is half the 3dB beamwidth, $G_t^{\max}$ is the maximum gain of the antenna in the main lobe and depends on the antenna diameter D_s , and carrier frequency f . Here, L_N , L_f , z_s , α_s , a_s , and b_s are antenna model parameters. The received signal at user k served by satellite i at time step t , can be expressed as

$$y_{k,i}^t = s_{k,i}^t \sqrt{p_{k,i}^t} h_{k,i \rightarrow k}^t x_k^t + \sum_{i' \in \mathcal{M}, i' \neq i} \sum_{j \in \mathcal{K}, j \neq k} s_{j,i'}^t \sqrt{p_{j,i'}^t} h_{k,i' \rightarrow j}^t x_j^t + n_k^t, \quad (12)$$

where x_k^t is the transmitted signal for user k , n_k shows the noise at user which follows independent Gaussian distribution with zero mean and the variance σ^2 , and $p_{k,i}^t$ is the transmit power from satellite i to user k at time t . Accordingly, we obtain the signal-to-interference-plus-noise ratio (SINR) for k -th user from i -th satellite as follows,

$$\gamma_{k,i}^t(s_t, \mathbf{p}_t) = \frac{s_{k,i}^t p_{k,i}^t |h_{k,i \rightarrow k}^t|^2}{\sigma^2 + \sum_{i' \in \mathcal{M}, i' \neq i} \sum_{j \in \mathcal{K}, j \neq k} s_{j,i'}^t p_{j,i'}^t |h_{k,i' \rightarrow j}^t|^2}, \quad (13)$$

where $s_t \in \{0, 1\}^{1 \times KM}$, and $\mathbf{p}_t \in \mathbb{R}^{1 \times KM}$, denote the association and transmit power vectors for time t . We use $r_{k,i}^t$ to represent the spectral efficiency of the user k in the time interval t from satellite i and it is calculated as

$$r_{k,i}^t(s_t, \mathbf{p}_t) = \log_2(1 + \gamma_{k,i}^t(s_t, \mathbf{p}_t)). \quad (14)$$

We continue with the proposed codebook design and the formulated problem in the next section.

III. PROPOSED FRAMEWORK

A. Proposed Codebook Design

For the user terminal antenna, we consider a small UPA with N_x and N_y elements along the x- and y-axes, respectively, separated by spacings d_x and d_y , and employing electronic beam steering, as illustrated in Fig. 3 (a). The codebook $\mathbf{W} = [\mathbf{w}_0, \mathbf{w}_1, \dots, \mathbf{w}_L]$ is considered for our proposed array antenna, containing $L + 1$ codewords, where $l = 0, 1, \dots, L$ indexes each codeword $\mathbf{w}_l$. We divide the three-dimensional scan range into M_ω layers of beams covering the elevation plane, where ω denotes the zenith receiving angle, and each layer contains N_ψ directions covering the azimuth plane, with

ψ representing the azimuth angle, as illustrated in Fig. 3 (b), and Fig. 3 (c), respectively. Additionally, we add a broadside beam to complete the coverage. Each codeword corresponds to a specific beam direction, and combined, they provide full coverage across the azimuth and elevation scan range. The array elements have an important role in the gain and scanning range. A wider HPBW in the element pattern results in a lower peak gain but an increased scan range. In our design, a wide-angle element pattern is preferable, as it enables full coverage with a smaller set of pointing subspaces, resulting in a more compact and efficient codebook for tracking fast-moving LEO satellites. Each element radiation factor is modeled according to [33] with the element pattern denoted by A_e as

$$\begin{aligned} A_e(\omega, \psi) &= G_e^{\max} - \min\{-(A_{e,x}(\omega) + A_{e,y}(\psi)), A_m\}, \\ A_{e,x}(\omega) &= -\min\left\{12 \left(\frac{\omega}{\omega_{3dB}^e}\right)^2, SLA\right\}, \\ A_{e,y}(\psi) &= -\min\left\{12 \left(\frac{\psi}{\psi_{3dB}^e}\right)^2, A_m\right\}, \end{aligned} \quad (15)$$

where $G_e^{\max}$ represents the maximum directional gain of the element, while $A_{e,x}$ and $A_{e,y}$ denote its x- and y-axes radiation patterns, respectively. The parameters ψ_{3dB}^e and ω_{3dB}^e correspond to the x and y 3 dB beamwidths in the receiving angles ψ and ω , A_m defines the front-back ratio, and SLA specifies the sidelobe level limit. The resulting antenna radiation pattern, $A(\mathbf{w}_l, \omega, \psi)$, for the l -th codeword, is obtained by adding the element radiation with the array factor as follows,

$$\begin{aligned} A(\mathbf{w}_l, \omega, \psi) &= A_e(\omega, \psi) \\ &+ 10 \log_{10} \left(\left| \sum_{m=1}^{N_x} \sum_{n=1}^{N_y} u_{m,n}^l \cdot v_{m,n} \right|^2 \right), \end{aligned} \quad (16)$$

where $u_{m,n}^l$ is the weighting factor for codeword $\mathbf{w}_l$ and $v_{m,n} = \exp(j \frac{2\pi}{\lambda} (d_x(m-1) \sin \omega \cos \psi + d_y(n-1) \sin \omega \sin \psi))$ is the superposition vector of the planar array. The beam vector $\mathbf{w}_l$ has a coverage angle region expressed as

$$\mathcal{C}(\mathbf{w}_l) = \left\{ (\omega, \psi) \mid A(\mathbf{w}_l, \omega, \psi) > \max_{(\hat{\omega}, \hat{\psi})} A(\mathbf{w}_l, \hat{\omega}, \hat{\psi}) - \rho \right\}, \quad (17)$$

where $\rho > 0$ is a factor that enlarges the coverage region as it increases, with a 3 dB coverage achieved at $\rho = 3$. To achieve a full scan range, the combined coverage of all designed codeword beams must span the entire azimuth and zenith angle domain, i.e.,

$$\bigcup_{l=0}^L \mathcal{C}(\mathbf{w}_l) = \left\{ (\omega, \psi) \mid [0, \omega_{\max}] \times [-\pi, \pi) \right\}, \quad (18)$$

where $\omega_{\max}$ represents the maximum zenith angle that determines the minimum elevation angle $\beta_{\min}$, given by $\beta_{\min} = \pi/2 - \omega_{\max}$. Each designed codeword has the coverage area as

$$\begin{aligned} \mathcal{C}(\mathbf{w}_l) &= \left\{ (\omega, \psi) \mid \left[\omega - \frac{B_\omega}{2}, \omega + \frac{B_\omega}{2} \right], \left[\psi - \frac{B_\psi}{2}, \psi + \frac{B_\psi}{2} \right] \right\}, \end{aligned} \quad (19)$$

which means that the resulting steering beam has the beam width B_ω and B_ψ for the elevation and azimuth, respectively. Here, we assume similar beamwidths for all codewords for notational simplicity, although off-boresight beams are slightly wider in practice. We present the broadside beam using the codeword defined as

$$\mathcal{C}(\mathbf{w}_{l=0}) = \left\{ (\omega, \psi) \mid \left[\omega - \frac{B_\omega}{2}, \omega + \frac{B_\omega}{2} \right], \psi = 0 \right\}. \quad (20)$$

For the rest of the beams, as previously mentioned, the elevation angle is divided into M_ω layers, each compromising N_ψ azimuth directions. The corresponding codewords are as

$$\begin{aligned} \mathcal{C}(\mathbf{w}_{l=1,\dots,L}) &= \left\{ (\omega, \psi) \mid \left[\frac{B_\omega}{2}(2m_c - 3), \frac{B_\omega}{2}(2m_c - 1) \right], \right. \\ &\quad \left. \left[\frac{B_\psi}{2} \left(n_c - 1 - \frac{N_\psi}{2} \right), \frac{B_\psi}{2} \left(n_c + 1 - \frac{N_\psi}{2} \right) \right] \right\}, \\ m_c &= 1, 2, \dots, M_\omega, \quad n_c = 1, 2, \dots, N_\psi, \end{aligned} \quad (21)$$

where $L = M_\omega N_\psi$ and $l = (m_c - 1)M_\omega + n_c$. With the array model established featuring $L + 1$ beam steering directions, we now aim to determine the optimal receiving codeword for each specific satellite. The receiving angle from satellite i for each l -th receiving pattern is

$$\begin{aligned} \theta_{i,k}^t(l) &= \arccos \left(\sin(\omega_{i,k}^t) \sin(\omega_l) \right. \\ &\quad \left. + \cos(\omega_{i,k}^t) \cos(\omega_l) \cos(\psi_{i,k}^t - \psi_l) \right), \end{aligned} \quad (22)$$

where we have, $\Theta_{i,k}^t = \{\theta_{i,k}^t(0), \theta_{i,k}^t(1), \dots, \theta_{i,k}^t(L+1)\}$ as the set of these receiving angles. The user terminal shall select the best codeword based on the closest maximum antenna gain to the i -th satellite. Accordingly, we can find the index of the fitting codeword by

$$\hat{l}_{i,k}^t = \underset{l}{\operatorname{argmin}} (\Theta_{i,k}^t). \quad (23)$$

The receiver antenna gain at user k , for the satellite i is obtained following equation (16) as

$$\operatorname{Gr}(\omega_{i,k}^t, \psi_{i,k}^t) = A(\mathbf{w}_{\hat{l}_{i,k}^t}, \omega_{i,k}^t, \psi_{i,k}^t). \quad (24)$$

In this way, for each candidate satellite i that may be associated with user k , the best codeword can be directly determined.

B. Problem Formulation

Our objective is to associate users with satellites while maintaining a defined QoS level and ensuring interference-free coexistence. To guarantee the required QoS, we enforce a minimum spectral efficiency threshold at the user, expressed as

$$\sum_{i \in \mathcal{M}} r_{k,i}^t(\mathbf{s}_t, \mathbf{p}_t) \geq R_{\min}^k, \quad (25)$$

where $R_{\min}^k$ denotes the minimum required spectral efficiency of user k . We formulate the dynamic user association in the coexisting NGSO-GSO satellite network as an optimization problem with handover minimization. The optimal user association and beam power are found in the time window T . We formulate the HOMN problem as follows,

$$\begin{aligned} \mathcal{P}_{\text{HOMN}} : \quad & \min_{\mathbf{S}, \mathbf{P}} \sum_{k \in \mathcal{K}} \sum_{i \in \mathcal{M}} \sum_{t \in \mathcal{T}} |s_{k,i}^t - s_{k,i}^{t-1}| / 2 \\ \text{s.t. } C_1 : & \sum_{i \in \mathcal{M}} r_{k,i}^t(\mathbf{s}_t, \mathbf{p}_t) \geq R_{\min}^k, \quad \forall t, k, \\ C_2 : & \sum_{i \in \mathcal{M}} s_{k,i}^t \leq 1, \quad \forall k, t, \\ C_3 : & \sum_{k \in \mathcal{K}} s_{k,i}^t \leq b_{\max}, \quad \forall t, i, \\ C_4 : & s_{k,i}^t \left(\frac{\beta_{\min}}{\beta_{k,i}} \right) \leq 1, \quad \forall t, k, i, \\ C_5 : & \sum_{k \in \mathcal{K}} s_{k,i}^t \leq 1 - FZ_{k,i}^t, \quad \forall t, i, \\ C_6 : & 0 \leq p_{k,i}^t \leq s_{k,i}^t P_{\max}, \quad \forall t, k, i, \\ C_7 : & s_{k,i}^t \in \{0, 1\}, \quad \forall k, i, t, \end{aligned} \quad (26)$$

where $\mathbf{S} \in \{0, 1\}^{T \times K \times M}$ denotes the overall association matrix, $\mathbf{P} \in \mathbb{R}^{T \times K \times M}$ the transmit power matrix, with $k \in \mathcal{K}$, $i \in \mathcal{M}$, and $t \in \mathcal{T}$, while $P_{\max}$ and $b_{\max}$ represent the maximum beam power and the maximum number of satellite beams, respectively. Here, $\beta_{k,i}$ represents the elevation angle of satellite i from the perspective of user k , and $\beta_{\min}$ denotes the minimum required elevation angle for satellite visibility (recall $\beta_{\min} = \pi/2 - \omega_{\max}$ from Section III-A). It should be noted that the user-to-satellite association inherently accounts for the codebook-based receive antenna gain, since the codebook design is embedded in the channel model and directly affects the rate calculation of C_1 .

The objective function minimizes the handover of all users throughout time T . Constraint C_1 insures the required spectral efficiency for each user at each time step is satisfied. C_2 confines that each user is not assigned to more than one satellite. The number of users per satellite is limited to $b_{\max}$ beams by constraint C_3 , and C_4 guarantees that the elevation angle of the assigned satellite, satisfies user's minimum threshold. C_5 guarantees the interference protection to GSO services by avoiding assigning users to the satellites which have the forbidden zones in the area of the user. The satellite power limit per user is defined in C_6 . Finally, C_7 indicates that association indicators $s_{k,i}^t$ are binary variables.

The optimization problem $\mathcal{P}$ is challenging due to several sources of non-convexity, which are summarized as follows:

- The user-to-satellite association variables are binary and time-dependent, introducing integer constraint in C_7 that makes the problem a mixed-integer nonlinear programming (MINLP) problem.
- The user SINR and spectral efficiency constraint C_1 involves bilinear coupling between association variables and transmit power variables, resulting in non-convex terms.
- The C_1 constraint also includes inter satellite interference, leading to a fractional and non-convex constraint structure that depends jointly on the association and power allocations of multiple users and satellites.

Due to these challenges, the resulting problem is a non-convex MINLP which is mathematically intractable and computationally hard. It is important to note that finding an optimal solution is challenging, and conventional methods for solving convex optimization problems cannot be directly applied. In the following we present our proposed optimization solution.

IV. PROBLEM SOLUTION

First we consider the non-convex spectral efficiency constraint C_1 , and rewrite it as

$$\sum_{i \in \mathcal{M}} \log_2 (1 + \gamma_{k,i}^t(\mathbf{s}_t, \mathbf{p}_t)) \geq R_{\min}^k. \quad (27)$$

By substituting equation (13) in the left-hand side of the above inequality, we obtain

$$\begin{aligned} & \sum_{i \in \mathcal{M}} \log_2 \left(1 + \frac{s_{k,i}^t p_{k,i}^t |h_{k,i \rightarrow k}^t|^2}{\sigma^2 + \sum_{i' \in \mathcal{M}, i' \neq i} \sum_{j \in \mathcal{K}, j \neq k} s_{j,i'}^t p_{j,i'}^t |h_{k,i' \rightarrow j}^t|^2} \right) \\ & \geq R_{\min}^k. \end{aligned} \quad (28)$$

In this inequality, we have the multiplication of two variables which is not desirable. We can simply remove the variable s from both the nominator and denominator since with constraint C_6 , if the satellite is not assigned to the user, i.e., $s_{k,i}^t = 0$, the power variable will be zero as well. Accordingly, the constraint is simplified as

$$\begin{aligned} & \sum_{i \in \mathcal{M}} \log_2 \left(1 + \frac{p_{k,i}^t |h_{k,i \rightarrow k}^t|^2}{\sigma^2 + \sum_{i' \in \mathcal{M}, i' \neq i} \sum_{j \in \mathcal{K}, j \neq k} p_{j,i'}^t |h_{k,i' \rightarrow j}^t|^2} \right) \\ & \geq R_{\min}^k. \end{aligned} \quad (29)$$

We restate it as

$$\sum_{i \in \mathcal{M}} (\mathbf{r}_1(\mathbf{p}_t) - \mathbf{r}_2(\mathbf{p}_t)) \geq R_{\min}^k, \quad (30)$$

where

$$\mathbf{r}_1(\mathbf{p}_t) = \log_2 \left(p_{k,i}^t |h_{k,i \rightarrow k}^t|^2 \sum_{i' \in \mathcal{M}, i' \neq i} \sum_{j \in \mathcal{K}, j \neq k} p_{j,i'}^t |h_{k,i' \rightarrow j}^t|^2 + \sigma^2 \right), \quad (31)$$

and

$$\mathbf{r}_2(\mathbf{p}_t) = \log_2 \left(\sum_{i' \in \mathcal{M}, i' \neq i} \sum_{j \in \mathcal{K}, j \neq k} p_{j,i'}^t |h_{k,i' \rightarrow j}^t|^2 + \sigma^2 \right). \quad (32)$$

The constraint in (30) is the difference of convex functions and still non-convex. We can apply the first-order Taylor approximation for $\mathbf{r}_2(\mathbf{p}_t)$ around the $(f-1)$ -th iteration as follows,

$$\tilde{\mathbf{r}}_2(\mathbf{p}_t) = \mathbf{r}_2(\mathbf{p}_t^{f-1}) + \nabla_{\mathbf{p}} \mathbf{r}_2(\mathbf{p}_t^{f-1}) (\mathbf{p}_t - \mathbf{p}_t^{f-1}), \quad (33)$$

that is obtained as

$$\begin{aligned} \tilde{\mathbf{r}}_2(\mathbf{p}_t) = & \log_2 \left(\sum_{i' \in \mathcal{M}, i' \neq i} \sum_{j \in \mathcal{K}, j \neq k} (p_{j,i'}^t)^{f-1} (h_{k,i' \rightarrow j}^t)^2 + \sigma^2 \right) \\ & + \frac{1}{\ln 2} \frac{\sum_{i' \in \mathcal{M}, i' \neq i} \sum_{j \in \mathcal{K}, j \neq k} (p_{j,i'}^t - (p_{j,i'}^t)^{f-1}) |h_{k,i' \rightarrow j}^t|^2}{\sum_{i' \in \mathcal{M}, i' \neq i} \sum_{j \in \mathcal{K}, j \neq k} (p_{j,i'}^t)^{f-1} (h_{k,i' \rightarrow j}^t)^2 + \sigma^2}. \end{aligned} \quad (34)$$

Hence, the convexified QoS constraint is written as

$$C_{1a} : \sum_{i \in \mathcal{M}} (\mathbf{r}_1(\mathbf{p}_t) - \tilde{\mathbf{r}}_2(\mathbf{p}_t)) \geq R_{\min}^k, \quad \forall t, k. \quad (35)$$

Note that the multi-dimensional binary variable $\mathbf{s}$ still makes the problem $\mathcal{P}$ non-convex. We can relax the integer constraint C_7 and add a penalty function in the objective function to enforce the variables to take values of 0 or 1. The resulting problem is written as

$$\begin{aligned} \mathcal{P}'_{HOMN} : & \min_{\mathbf{s}, \mathbf{p}} \sum_{k \in \mathcal{K}} \sum_{i \in \mathcal{M}} \sum_{t \in \mathcal{T}} \left(\frac{|s_{k,i}^t - s_{k,i}^{t-1}|}{2} \right. \\ & \quad \left. + \mu (s_{k,i}^t - (s_{k,i}^t)^2) \right) \\ \text{s.t. } & C_{1a}, C_2, C_3, C_4, C_5, C_6, \\ & C_{7a} : 0 \leq s_{k,i}^t \leq 1, \quad \forall t, k, i, \end{aligned} \quad (36)$$

where constraint C_{7a} is the relaxed constraint C_7 , and μ is the penalty factor with a big value. It can be proven that $\mathcal{P}'$ is equivalent to problem $\mathcal{P}$ when the penalty factor is sufficiently large [34]. The objective function of the problem $\mathcal{P}'$ is still non-convex, since it is the difference of two convex functions. To resolve this we can rewrite the objective function as

$$\sum_{k \in \mathcal{K}} \sum_{i \in \mathcal{M}} \sum_{t \in \mathcal{T}} (H(\mathbf{S}) - \mu J(\mathbf{S})), \quad (37)$$

where

$$H(\mathbf{S}) = \frac{|s_{k,i}^t - s_{k,i}^{t-1}|}{2}, \quad (38)$$

and

$$J(\mathbf{S}) = (s_{k,i}^t)^2 - s_{k,i}^t. \quad (39)$$

Similar to the previous approach, we apply the first-order Taylor approximation for $J(\mathbf{S})$ and obtain

$$\tilde{J}(\mathbf{S}) = s_{k,i}^t (2(s_{k,i}^t)^{f-1} - 1) - ((s_{k,i}^t)^{f-1})^2, \quad (40)$$

Algorithm 1 : HOMN Method**Input:** System parameters and thresholds.**Output:** Optimal user-to-satellite association, $\mathbf{S}^*$, and transmit power, $\mathbf{P}^*$ matrices,

- 1: Find the set of satellites, $\mathcal{M}$,
- 2: Initialize variables for iteration $f = 0$, $(s_{k,i}^t)^f$, $(p_{k,i}^t)^f$, penalty factor $\mu \gg 1$,
- 3: **while** $f \leq \text{Iteration}_{max}$:
- 4: $f = f + 1$
- 5: Update $\tilde{\mathbf{r}}_2(\mathbf{p})$ from equation (33),
- 6: Update $\tilde{J}(\mathbf{S})$ from equation (40),
- 7: Solve problem $\mathcal{P}'$ from equation (41),
- 8: Update $(s_{k,i}^t)^f$, $(p_{k,i}^t)^f$,
- 9: **end while**
- 10: **return:** $\mathbf{S}^* = \mathbf{S}^f$, $\mathbf{P}^* = \mathbf{P}^f$.

where $(s_{k,i}^t)^{f-1}$ is the value for $(f-1)$ -th iteration. Finally, we obtain the convexified optimization problem as follows,

$$\begin{aligned} \mathcal{P}'_{HOMN} : \min_{\mathbf{S}, \mathbf{P}} \quad & \sum_{k \in \mathcal{K}} \sum_{i \in \mathcal{M}} \sum_{t \in \mathcal{T}} (H(\mathbf{S}) - \mu \tilde{J}(\mathbf{S})) \\ \text{s.t.} \quad & C_{1a}, C_2, C_3, C_4, C_5, C_6, C_{7a}. \end{aligned} \quad (41)$$

The objective function and the problem constraints are all convex and the optimized solution can be obtained by conventional solvers such as CVX [35]. Our proposed solution for the HOMN method is presented in Algorithm 1. The computational complexity of this algorithm is concentrated in the 7th step, which involves solving problem $\mathcal{P}'$ from (41). We can approximate the order of computation complexity as $\mathcal{O}(L_{itr} \sqrt{T(2K + 2M + 3KM)}(2KMT)^3)$, where L_{itr} presents the number of iterations required for the algorithm [36].

V. SIMULATION RESULTS

In the simulation setup, we assume the LEO satellites are from the double-shell constellation whose characteristics are presented in Table II. The system parameters considered in the simulations are listed in Table III. To compare our proposed handover method, HOMN, we have considered three benchmark methods.

- **Lnm**, link quality maximization handover, adapted from the conditional handover mechanism defined in 3GPP terrestrial standards and further studied for the non-terrestrial networks [38], where the decision is based on the reference signal received power (RSRP). The Lnm initiates a handover when a target satellite is available with a received power that exceeds that of the serving satellite by a defined hysteresis margin. This serves as a practical baseline that makes decisions based on the current signal conditions.
- **Evm**, minimum elevation angle handover that initiates whenever the serving satellite's elevation angle falls below the minimum required threshold. The user then will be associated to the satellite with the highest elevation angle.

TABLE III
SIMULATION PARAMETERS

Parameter	Value
Carrier frequency	11.7GHz
Satellite transmitter antenna model	[32]
Transmitter antenna diameter, D_s	0.8m
Transmitter antenna losses, L_N, L_f	-30dB, 0dBi [32]
Transmitter antenna parameters, z_s, α_s, a_s, b_s	1, 1.5, 2.58, 6.32 [32]
Beam bandwidth	100MHz
Maximum beam power, P_{max}	2dBW
Satellite's beam HPBW	4°
Latitude of GSO satellites	0°
Diameter of GSO antenna terminal	0.7m [37]
Array elements on x, y -axes, N_x, N_y	4, 4
Element spacing, $d_x = d_y$	0.45λ
Element gain	8dBi [33]
Element HPBW, $\phi_{3dB}^e, \theta_{3dB}^e$	80°, 80° [33]
Element front-back ratio, A_m	30dB
Element sidelobe level, SL_{Av}	30, 30dB
Codebook elevation layers, N_ψ	2
Codebook azimuth sections, M_ω	12
Codeword elevation HPBW, B_ψ	24°
Codeword azimuth HPBW, B_ω	30°
Minimum elevation, β_{min}	30°
Noise temperature	260K

TABLE IV
ANGULAR COVERAGE OF CBS CODEWORDS

	$-\frac{\pi}{15} < \psi \leq \frac{\pi}{15}$	$\frac{\pi}{15} < \psi \leq \frac{\pi}{5}$	$\frac{\pi}{5} < \psi \leq \frac{\pi}{3}$
$-\pi < \omega \leq -\frac{5\pi}{3}$	$\mathcal{C}(\mathbf{w}_0)$	$\mathcal{C}(\mathbf{w}_1)$	$\mathcal{C}(\mathbf{w}_{13})$
$-\frac{5\pi}{3} < \omega \leq -\frac{2\pi}{3}$		$\mathcal{C}(\mathbf{w}_2)$	$\mathcal{C}(\mathbf{w}_{14})$
$\vdots$		$\vdots$	$\vdots$
$\frac{5\pi}{3} < \omega \leq \pi$		$\mathcal{C}(\mathbf{w}_{12})$	$\mathcal{C}(\mathbf{w}_{24})$

- **Rmx**, a sum rate maximization problem, when having the same constraints as (41), but the sum rate maximization as the objective function as

$$\begin{aligned} \mathcal{P}_{Rmx} : \max_{\mathbf{S}, \mathbf{P}} \quad & \sum_{k \in \mathcal{K}} \sum_{i \in \mathcal{M}} \sum_{t \in \mathcal{T}} r_{k,i}^t \\ \text{s.t.} \quad & C_{1a}, C_2, C_3, C_4, C_5, C_6, C_{7a}. \end{aligned} \quad (42)$$

We compare our algorithm with a link-based method, a geometry-based method (both ignoring interference constraint), and a sum-rate-based forbidden zone method, thereby providing a comprehensive evaluation of its effectiveness. Moreover, in our design, the user terminal is equipped with an antenna array of $N_x N_y = 16$ elements. The proposed CBS design has 25 codewords that covers the angle domains $[-\pi, \pi) \times [0, \pi/6]$, resulting in the user minimum elevation angle of $\beta_{min} = \pi/6$. The codewords and their coverage range is presented in Table IV. We compare our proposed CBS model with two alternative antenna beamforming approaches.

- **ABS**, exact beam steering toward the incoming signal, serving as the upper bound benchmark. The array weights are computed as, $u_{m,n} = \frac{1}{\sqrt{N_x N_y}} e^{j \frac{2\pi}{\lambda} (d_x(m-1) \sin \omega_s \cos \psi_s + d_y(n-1) \sin \omega_s \sin \psi_s)}$, where ω_s and ψ_s are the desired steering angles.
- **DBS**, a two dimensional DFT-based codebook of the same size as the CBS, where each codeword corresponds to one of the 25 DFT steering vectors.

This allows us to evaluate CBS against both a comparable codebook-based method, DBS, for a fair baseline, and an

TABLE V
COMPARISON OF BEAM STEERING METHODS AT LEO USER RECEIVER

Metric	ABS (Adaptive)	Codebook-based (Proposed)
Computational load	High (~456 for parameters in Table. III).	Very Low (~193 FLOPs for our proposed CBS).
Finite phase precision	Assumes infinite-precision phase shifters, later quantized to hardware state, leading to performance loss.	Inherently designed for finite-precision hardware.
Control overhead & latency	Per-element control; higher latency.	One index lookup; low latency.
Overall system	High-precision, computationally expensive, and hardware sensitive.	Robust, hardware-friendly, low complexity, and low-latency.

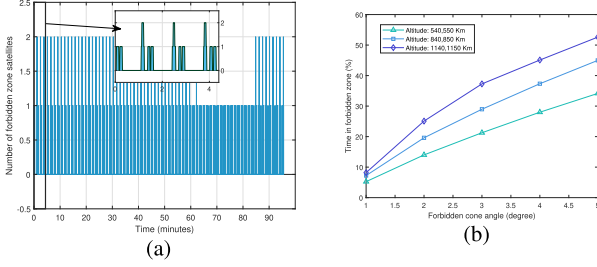


Fig. 4. (a) Number of satellites having forbidden zones at the location of the user during the constellation period. (b) Percentage of time the user falls in the forbidden zone of satellites for different cone angles and satellite altitudes.

idealized upper benchmark, ABS. The key operational and implementation differences between the ABS and our proposed codebook-based method are summarized in Table V. The comparison highlights the practical advantages of the CBS approach in terms of computational load and hardware compatibility, where floating point operations (FLOPs) are estimated per beam update based on the number of arithmetic operations¹ required to either compute beamforming weights (ABS) or select a predefined codebook (CBS).

Furthermore, the time step duration is set to $\Delta t = 4$ seconds, determined based on the rate of change in forbidden zones, which is influenced by the constellation parameters and the beamwidth of the satellites. We also assume uniform rate requirements for all users, setting $R_{\min}^k = R_{\min}$. In the following subsections, we evaluate the system's performance in terms of both interference and handover efficiency.

A. Interference Evaluation

In this section, we evaluate the occurrence of forbidden zones and their direct impact on user-to-satellite association and handover behavior. Since the proposed HOMN framework explicitly restricts user association when a user falls within a satellite's forbidden zone, the frequency and duration of these events directly affect the set of feasible serving satellites and, consequently, the handover rate. We consider a single user and evaluate the time instances during which this user falls in the forbidden zones of passing LEO satellites over one full constellation period. Forbidden zones are evaluated only for satellites that satisfy the user's minimum elevation angle requirement of 30 degrees. Note that since we have two orbital shells for the satellites, the simulation duration is chosen to cover the longer orbital period. We assume that the user is located at $(1.5^\circ, 16.5^\circ)$ which denotes the latitude

and longitude degrees, respectively. Fig. 4 (a) illustrates the number of satellites that have forbidden zones where the user is located during the constellation period time. We can see that there are periods where the user simultaneously is in the forbidden zone of two satellites, making them unavailable to serve the user. This reduces the pool of possible candidate satellites and may trigger handovers and force them to less favorable alternatives, directly motivating the handover minimization formulation adopted in our optimization framework. We further investigate the effect of the forbidden cone angle on the final restricted interference zones throughout the constellation period time. For comparison, we also consider a scenario in which the constellation operates at a higher altitude, specifically 1,150 km and 1,140 km for the first and second shells, respectively. The results for a user at the same location as before are shown in Fig. 4 (b). As the cone angle or satellite altitude increases, the projected footprint on the Earth grows, leading to a higher probability of the user falling within forbidden zones. This further emphasizes the necessity of the proposed HOMN method for joint handover and interference aware satellite selection.

We assess the proposed user-to-satellite association method by evaluating its interference on GSO terminals, using the ITU-defined equivalent power flux-density (EPFD) metric for NGSO compliance [39]. For a given GSO receiver, the EPFD is computed as

$$\text{EPFD} = 10 \log_{10} \left(\sum_{i \in \mathcal{V}} \frac{p_i \text{Gt}(\eta_i^G) G_r^G(\phi_i^G)}{4\pi d_i^{G^2} B_{ref} G_{rMax}^G} \right), \quad (43)$$

where $\mathcal{V}$ denotes the set of LEO satellites visible to the GSO receiver, p_i is the transmit power of LEO satellite i , $\text{Gt}(\eta_i^G)$ is the LEO satellite antenna gain toward the GSO receiver, $G_r^G(\phi_i^G)$ is the GSO earth station receive antenna gain with the maximum gain of G_{rMax}^G , d_i^G is the propagation distance, and B_{ref} is the reference bandwidth. In accordance with ITU, the GSO receiver antenna is modeled using the reference radiation patterns specified in [37], considering a small antenna size to account for consumer very small aperture terminals (VSATs). By ensuring protection for this reference envelope, which serves as a worst-case peak envelope, it provides a safety margin for the vast majority of operational GSO receivers.

We compare the user-to-satellite association under different handover methods and evaluate both the resulting interference toward coexisting GSO systems and the corresponding SINR at the user receiver, as shown in Fig. 5. The figure shows the assigned satellites for one representative LEO user under CBS, in a system with 5 users colocated with GSO terminals,

¹ Assuming add/mul: 1 FLOP, sin/cos: 10 FLOPs, $e^{j\theta}$: 20 FLOPs.

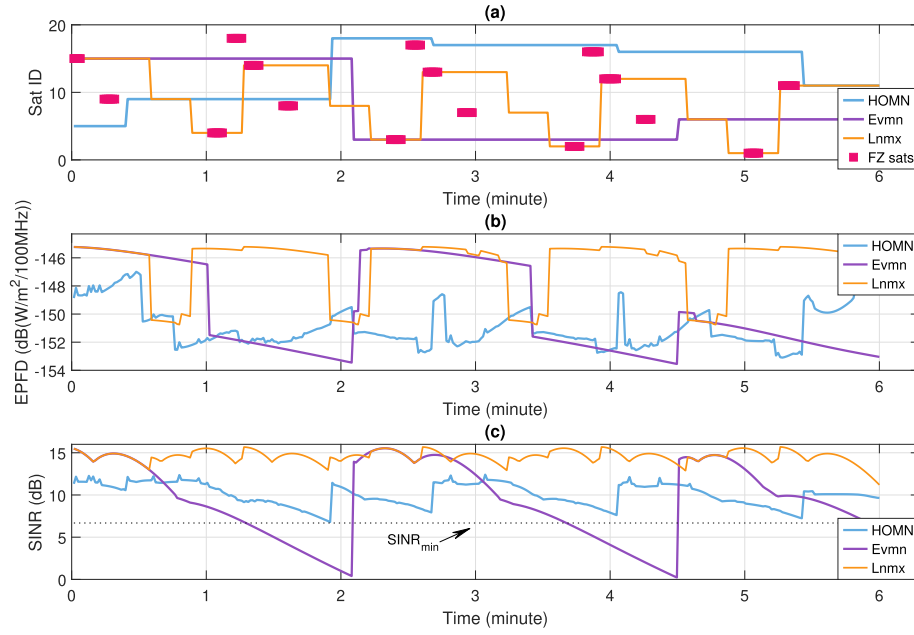


Fig. 5. Performance evaluation for users with CBS under $R_{\min} = 2.5$ bps/Hz and $T = 360$ s. (a) User-to-satellite association with the forbidden satellites (FZ sats), (b) LEO user SINR, and (c) EPFD levels at a co-located GSO ground terminal.

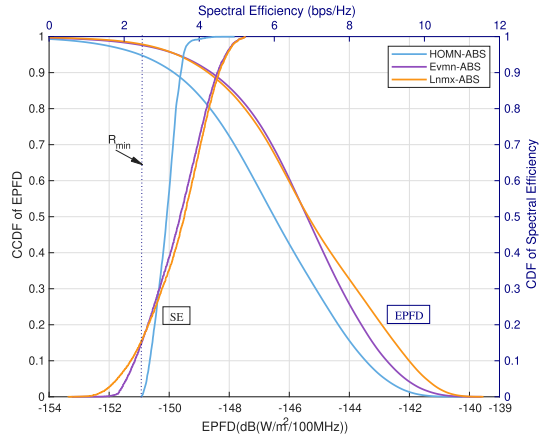


Fig. 6. Probabilities of EPFD interference at random GSO users and spectral efficiencies of the users for $R_{\min} = 2.4$ bps/Hz.

over $T = 6$ minutes. The minimum spectral efficiency is set to $R_{\min} = 2.5$ bps/Hz, corresponding to a SINR threshold of 6.68dB. Satellites restricted due to interference risk, based on their position relative to the user, are indicated with red squares. The results demonstrate that our HOMN method consistently associates the user with interference-free satellites while ensuring the required QoS. Compared to Evmn and LnmX, the proposed HOMN scheme reduces the average aggregate interference power at GSO receivers by 25% and 64%, respectively. In contrast, the Evmn and LnmX methods assign satellites without accounting for interference, resulting in higher EPFD levels at the GSO receiver. Moreover, Evmn fails to meet the minimum SINR requirement, as it is triggered only by the elevation angle and disregards link quality.

To further evaluate NGSO–GSO interference, we consider 2000 randomly distributed GSO terminal locations and compute the EPFD generated by the LEO constellation. Fig. 6 shows the CCDF of EPFD together with the CDF of spectral

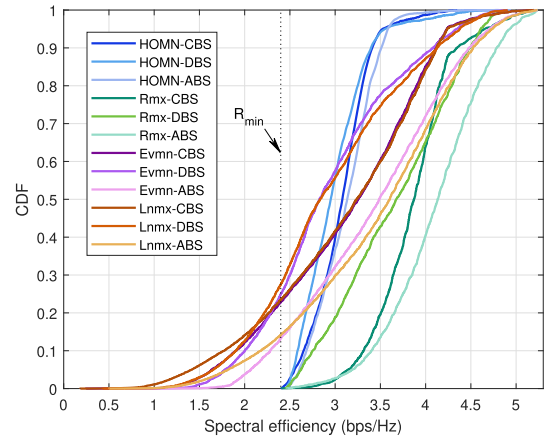


Fig. 7. CDF of users' spectral efficiency probabilities for $R_{\min} = 2.4$ bps/Hz.

efficiency for $K = 40$ LEO user terminals employing ABS, with $R_{\min} = 2.4$ bps/Hz and $T = 2$ minutes. The results reveal that the benchmark methods lead to a higher probability of elevated interference levels, while our proposed HOMN method effectively suppresses harmful interference. At an EPFD threshold of -142 dBW/m²/100MHz, HOMN reduces the probability of high interference events by 61.5% relative to Evmn and by 83.3% relative to LnmX. Furthermore, the spectral efficiency of HOMN consistently meets the minimum rate requirement, whereas the benchmark methods fail to do so in approximately 15% of cases.

B. Handover Evaluation

We evaluate performance using the handover rate, defined as $\mathcal{R}_{HO} = \frac{\mathcal{HO}}{K}$, where $\mathcal{HO}$ is given by equation (8). The results in this section are obtained over a duration of $T = 2$ minutes, considering $K = 40$ user terminals with $R_{\min} = 2.4$ bps/Hz. Each scenario is evaluated across 100

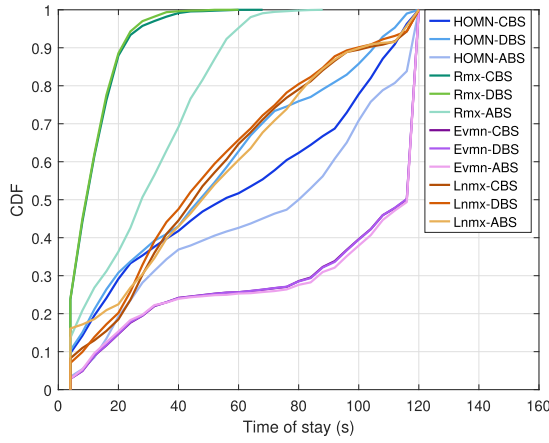


Fig. 8. CDF of users' time of stay probabilities for $R_{\min} = 2.4$ bps/Hz and $T = 120$ s.

iterations with randomly generated user locations. Fig. 7 presents the CDF of users' spectral efficiency under different handover strategies and beam steering models. As expected, the Rmx method achieves the highest spectral efficiencies, since its objective directly maximizes the network sum rate. However, as we later show, this comes at the cost of higher handover rates, which makes it less practical. The two baseline methods, Evmn and Lnmx, fail to consistently satisfy the minimum spectral efficiency requirement. Across all handover scenarios, the ABS receiver model serves as an upper bound due to its idealized beam steering capability. Our proposed CBS method outperforms the DBS baseline, improving median spectral efficiency across all association methods with an average gain of approximately 10%. Moreover, the HOMN strategy exhibits the steepest CDF slope, reflecting a more consistent user experience and improved fairness, as a larger fraction of users achieve rates close to the average performance.

We also evaluate the duration for which users remain associated with a satellite, referred to as the time of stay. The CDF of the metric is shown in Fig. 8. The Evmn method yields the longest time of stay since a handover is triggered only when the minimum elevation constraint is violated, representing the upper bound of this metric in the simulations. The figure also highlights the effectiveness of the proposed HOMN scheme in prolonging user associations and reducing handovers. When CBS is employed, HOMN achieves a median time of stay approximately six times higher than Rmx and about 28% higher than Lnmx, demonstrating significantly improved link stability. In comparison, the Lnmx method exhibits a shorter time of stay than Rmx, as both follow the best link strategy, but Lnmx introduces a margin that triggers later handovers to stronger links.

We now assess the performance of our user association method in terms of handover rate, user spectral efficiency, and satellite beam power, considering only the optimization-based approaches with minimum spectral efficiency constraints. Fig. 9 shows the results for the handover rate versus the minimum demand for the spectral efficiency of users. Our proposed HOMN association method effectively reduces handover occurrences, achieving an average reduction of 91.09% in handover rates compared to RMX for the CBS beam

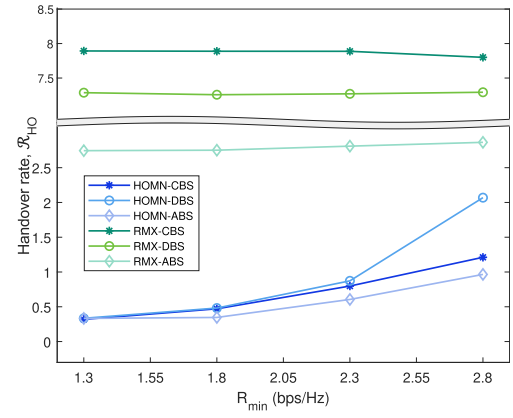


Fig. 9. Average handover rate for different minimum spectral efficiency demands.

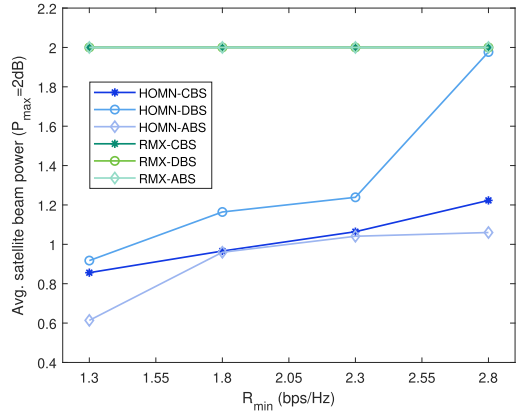


Fig. 10. Average satellite beam power for different minimum spectral efficiency demands.

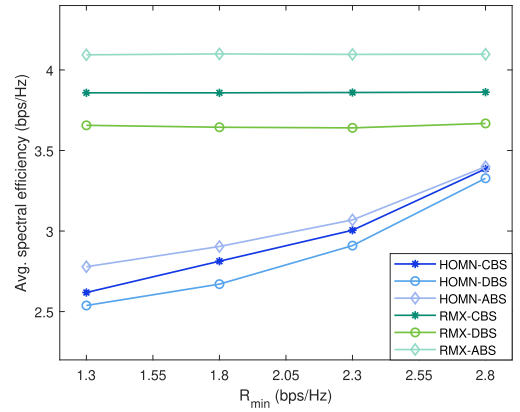


Fig. 11. Average users' spectral efficiency over different minimum demands.

steering method. As expected, the RMX approach results in a significantly higher handover rate, particularly when using the codebook-based beam steering methods. In this case, the absence of beam steering alignment to maximize received gain necessitates more frequent handovers to compensate for the gain loss. Furthermore, in the case of HOMN method, the DBS scheme results in more handovers than the proposed CBS approach, with the gap becoming especially pronounced at high values of $R_{\min}$.

The corresponding beam power values and spectral efficiencies are plotted in Fig. 10 and Fig. 11, respectively. The power efficiency of HOMN is evident in the results. Although the

CBS method requires slightly higher transmit power due to its receive gain limitations, its average increase over $R_{\min}$ values remains within 11.83% compared to the ABS model. Under the HOMN method, DBS requires on average 44% more transmit power than our CBS approach, highlighting the superior efficiency of the proposed codebook design. Moreover, we can see the maximum achievable spectral efficiencies for users with the RMX method in Fig. 11. These results are the upper bounds for spectral efficiency given the maximum satellite power gain usage. Despite its focus on minimizing handovers, our proposed method also enhances QoS while consuming less power.

VI. CONCLUSION

In this work, we proposed a user-to-satellite association and beam steering framework for NGSO satellite networks that ensures interference free coexistence with GSO systems while minimizing handovers. Our method dynamically identifies and avoids interference prone regions without requiring prior knowledge of the GSO user locations. To minimize the complexities associated with frequent handovers, the handover rate is integrated into the optimization objective. Moreover, we designed a low complexity codebook-based beam steering approach tailored for UPA antennas, enabling efficient beam scanning in the three dimensional angle domain. The performance evaluation demonstrates that the proposed HOMN scheme provides gain across multiple system-level metrics. In particular, HOMN significantly improves handover stability, achieving more than a fivefold increase in median time of stay compared to the benchmark RMX strategy, while also reducing handover rates by over 90%. In terms of interference management, HOMN achieves the lowest EPFD levels and suppresses high interference events, offering strong protection to coexisting GSO systems. Furthermore, the proposed CBS method reduces beam steering complexity and computational overhead, while retaining approximately 97% of the spectral efficiency of the ABS upper bound on average across different user spectral efficiency requirements, and consistently outperforming DBS in terms of spectral efficiency, handover rate, and power consumption.

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